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Space exploration has been the defining ambition guiding my career. JPL's Ingenuity mission captivated me from the moment I learned of it, and following its operations on Mars showed me how aerial robotics could transform planetary exploration. It directly shaped my decision to pursue a PhD focused on aerial Semantic SLAM, deliberately uniting my spacecraft and planetary-rover experience with deep, hands-on expertise in UAV autonomy. I now develop and deploy autonomous UAV systems across Semantic Relational SLAM, situational awareness, planning, and control using ROS 2, PX4, Linux, Docker, and embedded hardware. Earlier at Airbus Defence & Space, I developed rover GNC, LiDAR integration, and simulation software and supported time-constrained integrated breadboard campaigns. Together, this technical, systems, and team-leadership experience prepares me to help field reliable multi-agent aerospace robotic systems in challenging terrestrial and planetary environments.

Work Experience

Doctoral Researcher – Autonomous UAV Systems

November 2025 – Present

SnT, University of Luxembourg | Kirchberg, Luxembourg

- Develop Semantic Relational SLAM and situational-awareness architectures for UAVs, linking perception and semantic mapping with navigation, planning, and control.
- Integrate and deploy autonomy on physical UAVs using ROS 2, PX4, embedded Linux, Raspberry Pi, camera/IMU sensing, and OpenVINS visual-inertial odometry.
- Build reproducible Docker-based development, deployment, and camera-IMU calibration workflows with Kalibr, supported by version-controlled software and technical documentation.
- Conduct simulation, laboratory, hardware, and flight testing; troubleshoot interfaces across sensing, estimation, mapping, and autonomy to improve system reliability.
- Research collaborative UAV missions using shared semantic maps, resource-efficient navigation, and mission-aware multi-robot coordination.

AOCS/GNC Engineer Intern

July 2023 – August 2024

Airbus Defence & Space | Stevenage, UK

- Led LiDAR test activities during Integrated BreadBoarding (iBB) 3 & 4 campaigns, supporting an end-to-end Sample Fetch Rover concept demonstration at Technology Readiness Level 6.
- Designed and built a networked sensor test bench, performing low-level sensor integration by connecting LiDAR to rover breadboards using ROS over a local wireless network and analyzing campaign data.
- Developed C/C++ software on Linux for an offline GNC simulation, coupling a natural-scene generator to produce synthetic imagery and LiDAR scans; also investigated terrain generation and Vector Field Histogram planning for lunar rovers.
- Delivered software and hardware integration in a fast-paced Agile team using C/C++, Git, GitLab, Jira, and Confluence while working across GNC, simulation, and test disciplines.
- Worked directly with multidisciplinary engineers during time-constrained campaigns, communicating test status, integration issues, and technically grounded next steps.

Projects & Leadership

JamSail 3U CubeSat – AOCS/ADCS Lead

September 2024 – August 2025

NottSpace, University of Nottingham | Nottingham, UK

- Led ADCS design and implementation for a NottSpace 3U CubeSat created to test a GNSS-jamming sensing payload and a new solar-sail design, deriving subsystem requirements, architecture, interfaces, and a verification plan from the mission concept of operations.
- Developed embedded C flight software and a C/C++/CMake Linux simulation for attitude estimation and control, implementing an EKF that fuses sun-sensor, magnetometer, and IMU data alongside sensor filtering and control-state logic.
- Implemented the International Geomagnetic Reference Field (IGRF) in C, converting a mathematical Earth magnetic-field model into a physics-based input for simulation and flight-software development.
- Coordinated ADCS interfaces with the spacecraft team and applied ECSS-aligned software and verification practices across simulation and hardware testing.

Nottingham Space Society | Nottingham, UK

- Designed system architecture and subsystem interfaces within a student-led, 12-person Nottingham Space Society team developing a 3U CubeSat; led the requirements, Project Breakdown Tree, and Work Breakdown Structure.
- Developed software on a Raspberry Pi running Linux and integrated a microcontroller, IMU, and cameras before the concept progressed to the competition build phase.
- Contributed across embedded software, hardware and sensor integration, and system simulation while coordinating technical development within the student team.
- Achieved a top-four finish among 12 UK teams as first-time entrants to the UKSEDS Satellite Design Competition.
- Led cross-team meetings, communicated design changes, and presented the integrated concept to industry experts, balancing technical interfaces, build constraints, and competition deadlines.

UKSEDS Volunteer

Sept. 2024 – Sept. 2026

UK Students for the Exploration and Development of Space | Remote & UK field events

- Served as an Olympus Rover Trials (ORT) volunteer in the first year, helping organise a UK national rover competition with Airbus Defence & Space and RAL Space.
- Progressed in the second year to Competitions Management Co-Lead, overseeing four national competitions: ORT, SDC, ISAM, and NRC.
- Coordinated volunteers, schedules, technical milestones, participant support, and partner engagement with Airbus Defence & Space (SDC), the UK Space Agency (ISAM), Midlands Rocketry Club, and Skyrora (NRC).
- Produced clear participant documentation and supported venue and field-event logistics, helping multidisciplinary volunteer teams deliver competitions to fixed schedules.

Education & Training

MEng Aerospace Engineering – First-Class Honours

September 2020 – August 2025

University of Nottingham | Nottingham, UK

- Graduated with First-Class Honours, specializing in Aerospace Control Systems, Avionics, dynamics, software, modeling, and simulation.
- Applied C/C++, Python, MATLAB, Simulink, and Linux to estimation, control, flight dynamics, and autonomous-system projects.
- Completed design projects spanning aerospace systems, avionics, simulation, and control, providing the systems-engineering foundation for later space and robotics work.

IEEE RAS Summer School on Multi-Robot Systems

July 2026

Czech Technical University in Prague | Prague, Czechia

- Completed the 2026 IEEE RAS Summer School on Multi-Robot Systems, covering cooperative aerial robotics and active perception; presented the SaCoGraph collaborative-UAV research concept.
- Applied multi-robot autonomy concepts to collaborative sensing, semantic mapping, and resource-aware mission planning.

Space Resources & Robotics Summer School

May 2026

AGH University of Kraków | Krakow, Poland

- Completed intensive training in space resources, planetary robotics, and robotic manipulation for exploration applications.
- Developed a broader systems perspective on robotic operations in challenging off-world environments.

Space Systems Engineering Training Course

May 2025

ESA Academy | Transinne, Belgium

- Completed intensive ESA Academy training across the space-system lifecycle, from mission objectives and concepts of operations through architecture, integration, verification, and operations.
- Applied requirements engineering and concurrent-design methods to translate user needs into traceable, testable requirements and manage interfaces between technical disciplines.
- Practised system options and trade studies, budgets and margins, verification and validation planning, and the relationship between engineering decisions, schedule, cost, and risk.
- Completed multidisciplinary group exercises and communicated design rationale and systems-engineering deliverables, strengthening collaborative problem-solving across complex spacecraft projects.